

Primes and GCD

Subtitle

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OUTLINE

1. Primes and GCD

Primes

An integer p greater than 1 is called prime if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called composite.

Prime numbers: 2, 3, 5, 7, 11, 13, 17, ...

The Fundamental Theorem of Arithmetic

Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes, where the prime factors are written in order of nondecreasing size.

$$100 = 2.2.5.5 = 2^2.5^2$$

$$641 = 641$$

$$333 = 3.3.37 = 3^2.37$$

$$64 = 2.2.2.2.2.2 = 2^6$$

Theorem

If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

Example Show that 101 is prime.

Sol:

The only primes not exceeding $\sqrt{101}$ are 2, 3, 5, and 7. Because 101 is not divisible by 2, 3, 5, or 7 (the quotient of 101 and each of these integers is not an integer), it follows that 101 is prime.

Example Find the prime factorization of 7007.

Theorem

There are infinitely many primes.

Definition:

Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the greatest common divisor of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

Example: What is the greatest common divisor of 24 and 36?

Sol:

The positive common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12. Hence, $\gcd(24, 36) = 12$.

Example: What is the greatest common divisor of 17 and 22?

Sol:

The integers 17 and 22 have no positive common divisors other than 1, so that $\gcd(17, 22) = 1$.

Definition

The integers a and b are relatively prime if their greatest common divisor is 1.

Definition

The integers $a_1, a_2, \dots, a_n$ are pairwise relatively prime if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i \leq j \leq n$.

Example:

Determine whether the integers are pairwise relatively prime.

- a. 10, 17, and 21,
- b. 10, 19, and 24

Sol:

- a. Because $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$, we conclude that 10, 17, and 21 are pairwise relatively prime.
- b. Because $\gcd(10, 24) = 2 > 1$, we see that 10, 19, and 24 are not pairwise relatively prime.

gcd by prime factorisations.

Suppose that the prime factorizations of the positive integers a and b are:

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$

where each exponent is a nonnegative integer, and where all primes occurring in the prime factorization of either a or b are included in both factorizations, with zero exponents if necessary.

Then $\gcd(a, b)$ is given by:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \dots p_n^{\min(a_n, b_n)}$$

$\min(a_n, b_n)$ represents the minimum of the two numbers x and y .

Example:

Because the prime factorizations of 120 and 500 are

$$120 = 2^3 \cdot 3 \cdot 5 \text{ and}$$

$$500 = 2^2 \cdot 5^3, \text{ the}$$

$$\gcd(120, 500) = 2^{\min(3,2)} 3^{\min(1,0)} 5^{\min(1,3)} = 2^2 3^0 5^1 = 20$$

Definition:

The least common multiple (lcm) of the positive integers a and b is the smallest positive integer that is divisible by both a and b . The least common multiple of a and b is denoted by $\text{lcm}(a, b)$.

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \dots p_n^{\max(a_n, b_n)}$$

$\max(x, y)$ denotes the maximum of the two numbers x and y .

p_i its prime factors.

Example: What is the least common multiple of $2^3 3^5 7^2$ and $2^4 3^3$?

Sol: $\text{lcm}(2^3 3^5 7^2 \text{ and } 2^4 3^3) = 2^{\max(3,4)} 3^{\max(5,3)} 7^{\max(2,0)} = 2^4 3^5 7^2$.

Theorem

Let a and b be positive integers. Then
 $ab = \gcd(a, b) \cdot \text{lcm}(a, b).$

The Euclidean Algorithm

Lemma

Let $a = bq + r$, where a , b , q , and r are integers. Then $\gcd(a, b) = \gcd(b, r)$.

Example:

Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.

Sol:

Successive uses of the division algorithm give:

$$662 = 414 \cdot 1 + 248$$

$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41$$

Hence, $\gcd(414, 662) = 2$, because 2 is the last nonzero remainder.

Euclidean algorithm - tabular form

j	a	b	q	r
0	662	414	1	248
1	414	248	1	166
2	248	166	1	82
3	166	82	2	2
4	82	2	41	0

The End